

Machine Learning

Lecture 4: Linear Regression

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Winter term 2024/2025

Notation

Symbol	Meaning
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x	scalar is lowercase and not bold
$\boldsymbol{x}$	vector is lowercase and bold
$\boldsymbol{\Sigma}$	matrix is uppercase and bold
$f(\boldsymbol{x})$	predicted value for inputs $\boldsymbol{x}$
$\boldsymbol{y}$	vector of targets
y_i	target of the i 'th example
w_0	bias term (not to be confused with bias in general)
$\phi(\cdot)$	basis function
$E(\cdot)$	error function
$\mathcal{D}$	training data
$\boldsymbol{X}^\dagger$	Moore-Penrose pseudoinverse of $\boldsymbol{X}$

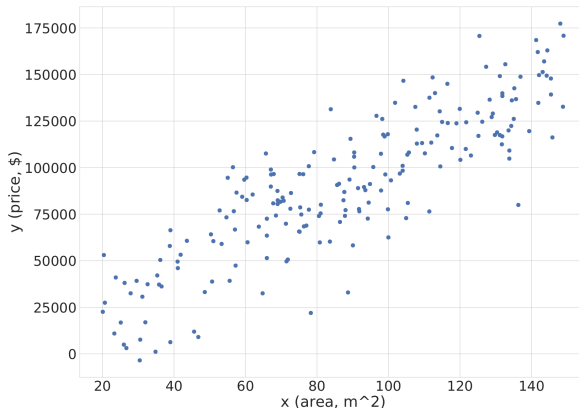
There is not a special symbol for vectors or matrices augmented by the bias term, w_0 . Assume it is always included.

Section 1

Basic Linear Regression

Example: Housing price prediction

Given is a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, of house areas x_i and corresponding prices y_i .



How do we estimate a price of a new house with area x_{new} ?

Regression problem

Given

- observations ¹

$$\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}, \mathbf{x}_i \in \mathbb{R}^D$$

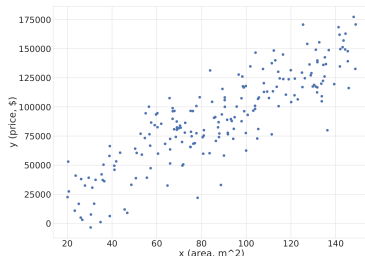
- targets

$$\mathbf{y} = \{y_1, y_2, \dots, y_N\}, y_i \in \mathbb{R}$$

Find

- Mapping $f(\cdot)$ from inputs to targets

$$y_i \approx f(\mathbf{x}_i)$$



¹A common way to represent the samples is as a **data matrix** $\mathbf{X} \in \mathbb{R}^{N \times D}$, where each row represents one sample.

Linear model

Target y is generated by a deterministic function f of $\mathbf{x}$ plus noise

$$y_i = f(\mathbf{x}_i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \beta^{-1}) \quad (1)$$

Let's choose $f(\mathbf{x})$ to be a linear function

$$f_{\mathbf{w}}(\mathbf{x}_i) = w_0 + w_1 x_{i1} + w_2 x_{i2} + \dots + w_D x_{iD} \quad (2)$$

$$= w_0 + \mathbf{w}^T \mathbf{x}_i \quad (3)$$

Absorbing the bias term

The linear function is given by

$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \dots + w_Dx_D \quad (4)$$

$$= w_0 + \mathbf{w}^T \mathbf{x} \quad (5)$$

Here w_0 is called **bias** or **offset** term. For simplicity, we can "absorb" it by prepending a 1 to the feature vector $\mathbf{x}$ and respectively adding w_0 to the weight vector $\mathbf{w}$:

$$\tilde{\mathbf{x}} = (1, x_1, \dots, x_D)^T \quad \tilde{\mathbf{w}} = (w_0, w_1, \dots, w_D)^T$$

The function $f_{\mathbf{w}}$ can compactly be written as $f_{\mathbf{w}}(\mathbf{x}) = \tilde{\mathbf{w}}^T \tilde{\mathbf{x}}$.

To unclutter the notation, we will assume the bias term is always absorbed and write $\mathbf{w}$ and $\mathbf{x}$ instead of $\tilde{\mathbf{w}}$ and $\tilde{\mathbf{x}}$.

From now we will always assume that the bias term is absorbed into the $\mathbf{x}$ vector

Loss function

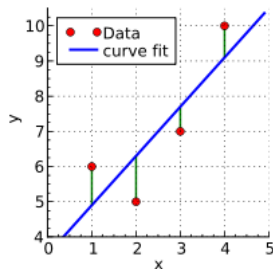
Now, how do we choose the "best" w that fits our data?

A **loss function** measures the "misfit" or error between our model (parametrized by w) and observed data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$.

Standard choice - **least squares** (LS)

$$E_{\text{LS}}(w) = \frac{1}{2} \sum_{i=1}^N (f_w(x_i) - y_i)^2 \quad (6)$$

$$= \frac{1}{2} \sum_{i=1}^N (w^T x_i - y_i)^2 \quad (7)$$



Factor $\frac{1}{2}$ is for later convenience

Objective

Find the optimal weight vector $\mathbf{w}^*$ that minimizes the error

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} E_{\text{LS}}(\mathbf{w}) \quad (8)$$

$$= \arg \min_{\mathbf{w}} \frac{1}{2} \sum_{i=1}^N (\mathbf{x}_i^T \mathbf{w} - y_i)^2 \quad (9)$$

By stacking the observations $\mathbf{x}_i$ as rows of the matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$

$$= \arg \min_{\mathbf{w}} \frac{1}{2} (\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y}) \quad (10)$$

Optimal solution

To find the minimum of the loss $E(\mathbf{w})$, compute the gradient $\nabla_{\mathbf{w}} E(\mathbf{w})$:

$$\nabla_{\mathbf{w}} E_{\text{LS}}(\mathbf{w}) = \nabla_{\mathbf{w}} \frac{1}{2} (\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y}) \quad (11)$$

$$= \nabla_{\mathbf{w}} \frac{1}{2} \left(\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{w}^T \mathbf{X}^T \mathbf{y} + \mathbf{y}^T \mathbf{y} \right) \quad (12)$$

$$= \mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{X}^T \mathbf{y} \quad (13)$$

See Equations (69), (81) from Matrix cookbook for details

Optimal solution

Now set the gradient to zero and solve for $\mathbf{w}$ to obtain the minimizer ²

$$\mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{X}^T \mathbf{y} \stackrel{!}{=} 0 \quad (14)$$

This leads to the so-called **normal equation** of the least squares problem

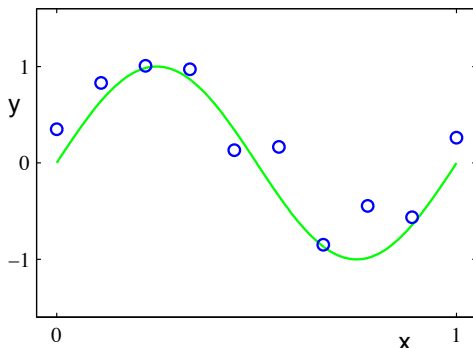
$$\mathbf{w}^* = \underbrace{(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T}_{=\mathbf{X}^\dagger} \mathbf{y} \quad (15)$$

$\mathbf{X}^\dagger$ is called **Moore-Penrose pseudo-inverse** of $\mathbf{X}$ (because for an invertible square matrix, $\mathbf{X}^\dagger = \mathbf{X}^{-1}$).

²Because Hessian $\nabla_{\mathbf{w}} \nabla_{\mathbf{w}} E(\mathbf{w})$ is positive (semi)definite $\rightarrow$ see *Optimization*

Nonlinear dependency in data

What if the dependency between y and x is not linear?



Data generating process: $y_i = \sin(2\pi x_i) + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \beta^{-1})$

For this example assume that the data dimensionality is $D = 1$

Polynomials

Solution: Polynomials are universal function approximators, so for 1-dimensional x we can define f as

$$f_{\mathbf{w}}(x) = w_0 + \sum_{j=1}^M w_j x^j \quad (16)$$

Or more generally

$$= w_0 + \sum_{j=1}^M w_j \phi_j(x) \quad (17)$$

Define $\phi_0 = 1$

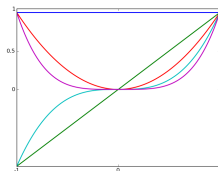
$$= \mathbf{w}^T \boldsymbol{\phi}(x) \quad (18)$$

The function f is still linear in $\mathbf{w}$ (despite not being linear in x)!

Typical basis functions

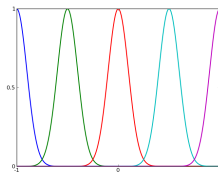
Polynomials

$$\phi_j(x) = x^j$$



Gaussian

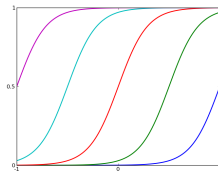
$$\phi_j(x) = e^{\frac{-(x-\mu_j)^2}{2s^2}}$$



Logistic Sigmoid

$$\phi_j(x) = \sigma\left(\frac{x-\mu_j}{s}\right),$$

where $\sigma(a) = \frac{1}{1+e^{-a}}$



Linear basis function model

For d -dimensional data $\mathbf{x}$: $\phi_j : \mathbb{R}^d \rightarrow \mathbb{R}$

Prediction for one sample

$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + \sum_{j=1}^M w_j \phi_j(\mathbf{x}) = \mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}) \quad (19)$$

Using the same least squares error function as before

$$E_{\text{LS}}(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^N (\mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i) - y_i)^2 = \frac{1}{2} (\boldsymbol{\Phi} \mathbf{w} - \mathbf{y})^T (\boldsymbol{\Phi} \mathbf{w} - \mathbf{y}) \quad (20)$$

with

$$\boldsymbol{\Phi} = \begin{pmatrix} \phi_0(\mathbf{x}_1) & \phi_1(\mathbf{x}_1) & \dots & \phi_M(\mathbf{x}_1) \\ \phi_0(\mathbf{x}_2) & \phi_1(\mathbf{x}_2) & & \vdots \\ \vdots & \vdots & \ddots & \\ \phi_0(\mathbf{x}_N) & \phi_1(\mathbf{x}_N) & \dots & \phi_M(\mathbf{x}_N) \end{pmatrix} \in \mathbb{R}^{N \times (M+1)}$$

being the **design matrix** of $\boldsymbol{\phi}$.

Optimal solution

Recall the final form of the least squares loss that we arrived at for the original feature matrix $\mathbf{X}$

$$E_{\text{LS}}(\mathbf{w}) = \frac{1}{2}(\mathbf{X}\mathbf{w} - \mathbf{y})^T(\mathbf{X}\mathbf{w} - \mathbf{y})$$

and compare it to the expression we found with the design matrix $\Phi \in \mathbb{R}^{N \times (M+1)}$

$$E_{\text{LS}}(\mathbf{w}) = \frac{1}{2}(\Phi\mathbf{w} - \mathbf{y})^T(\Phi\mathbf{w} - \mathbf{y}). \quad (21)$$

This means that the optimal weights $\mathbf{w}^*$ can be obtained in the same way

$$\mathbf{w}^* = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y} = \Phi^\dagger \mathbf{y} \quad (22)$$

Compare this to Equation 15:

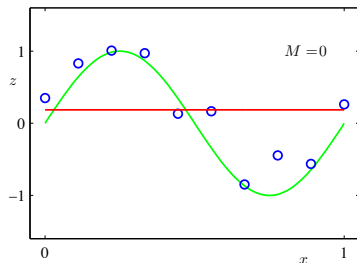
$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{X}^\dagger \mathbf{y} \quad (23)$$

Choosing degree of the polynomial

How do we choose the degree of the polynomial M ?

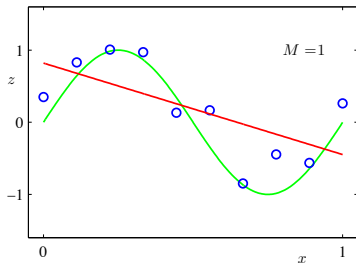
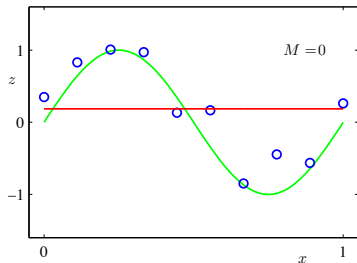
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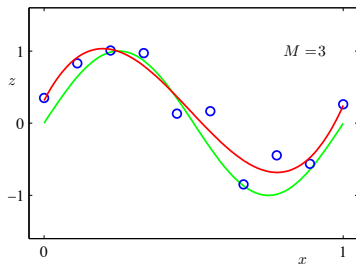
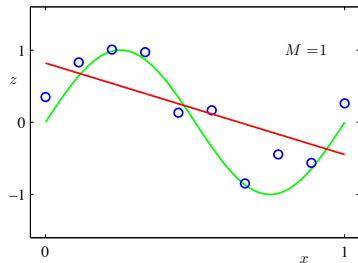
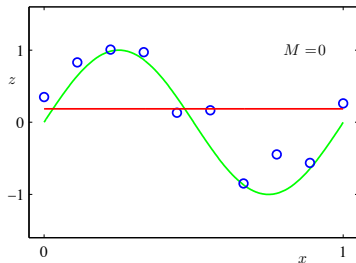
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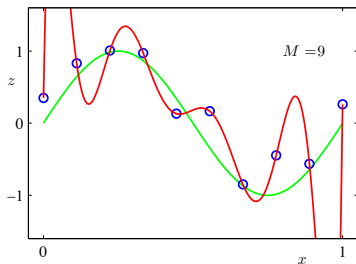
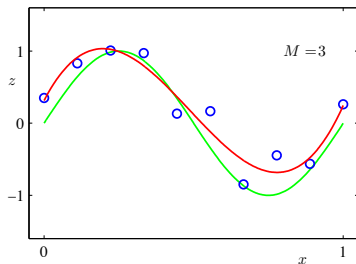
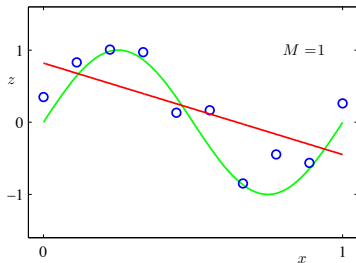
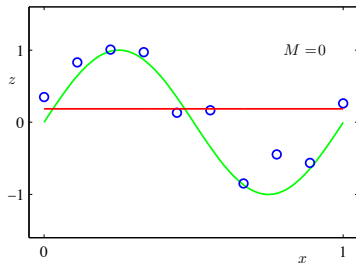
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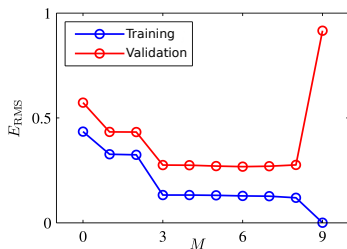
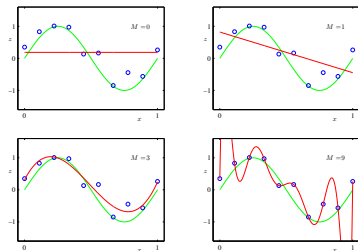


Choosing degree of the polynomial

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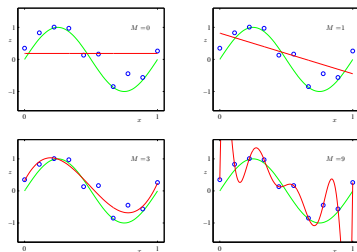


Choosing degree of the polynomial



One valid solution is to choose M using the standard train-validation split approach.

Choosing degree of the polynomial



	$M = 0$	$M = 1$	$M = 3$	$M = 9$
w_0^*	0.19	0.82	0.31	0.35
w_1^*		-1.27	7.99	232.37
w_2^*			-25.43	-5321.83
w_3^*			17.37	48568.31
w_4^*				-231639.30
w_5^*				640042.26
w_6^*				-1061800.52
w_7^*				1042400.18
w_8^*				-557682.99
w_9^*				125201.43

We also make another observation: overfitting occurs when the coefficients w become large.

What if we penalize large weights?

Controlling overfitting with regularization

Least squares loss with **L2 regularization** (also called **ridge regression**)

$$E_{\text{ridge}}(\mathbf{w}) = \frac{1}{2} \sum^N [\mathbf{w}^T \phi(\mathbf{x}_i) - y_i]^2 + \frac{\lambda}{2} \|\mathbf{w}\|_2^2 \quad (24)$$

where

- $\|\mathbf{w}\|_2^2 \equiv \mathbf{w}^T \mathbf{w} = w_0^2 + w_1^2 + w_2^2 + \dots + w_M^2$ - squared L2 norm of $\mathbf{w}$
- λ - regularization strength

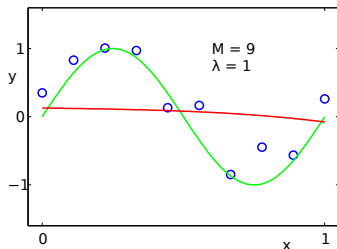
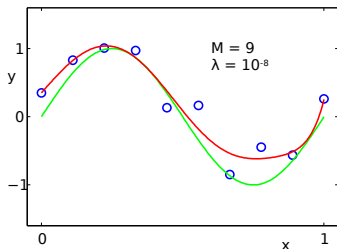
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- $\|\mathbf{w}\|_2^2 \equiv \mathbf{w}^T \mathbf{w} = w_0^2 + w_1^2 + w_2^2 + \dots + w_M^2$ - squared L2 norm of $\mathbf{w}$
- λ - regularization strength



Larger regularization strength λ leads to smaller weights $\mathbf{w}$

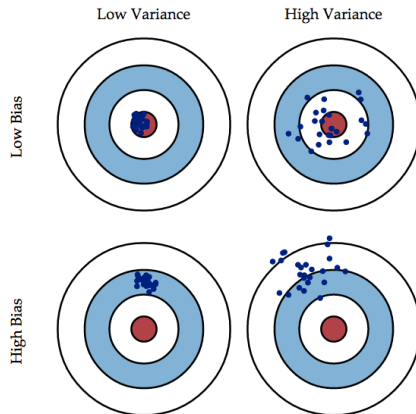
Bias-variance tradeoff

The error of an estimator can be decomposed into two parts: ³

- **Bias** - expected error due to model mismatch
- **Variance** - variation due to randomness in training data

the center of the target:
the true model that predicts
the correct values.

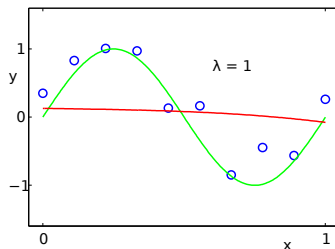
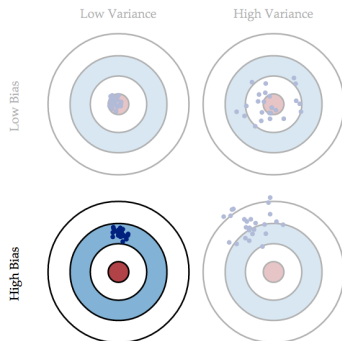
different hits (the blue dots):
different realizations of model
given different training data.



³See Bishop Section 3.2 for a more rigorous mathematical derivation

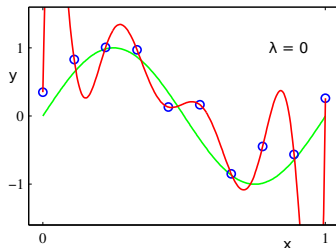
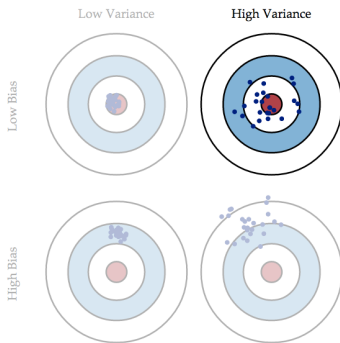
Bias-variance tradeoff: high bias

- In case of **high bias**, the model is too rigid to fit the underlying data distribution.
- This typically happens if the model is misspecified and/or the regularization strength λ is too high.



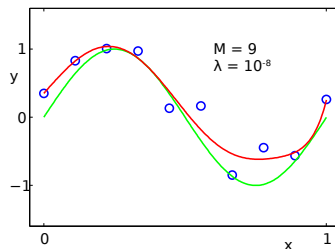
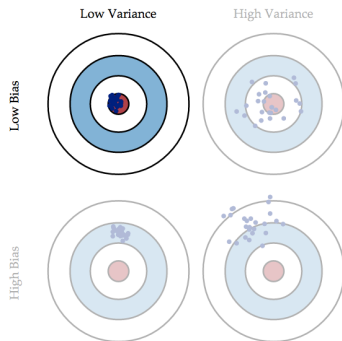
Bias-variance tradeoff: high variance

- In case of **high variance**, the model is too flexible, and therefore captures noise in the data.
- This is exactly what we call **overfitting**.
- This typically happens when the model has high capacity (= it "memorizes" the training data) and/or λ is too low.



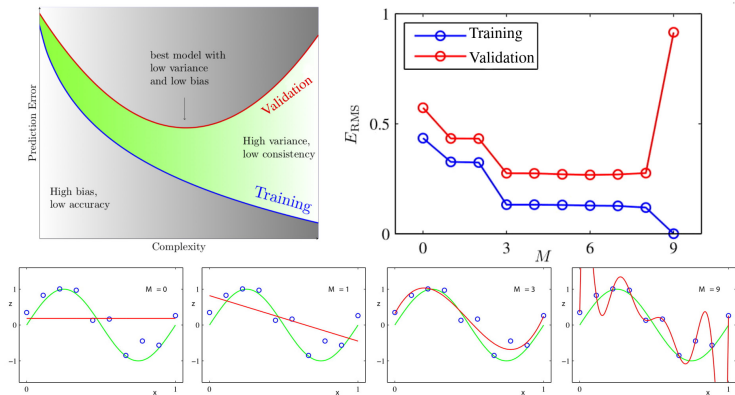
Bias-variance tradeoff

- Of course, we want models that have low bias and low variance, but often those are conflicting goals.
- A popular technique is to select a model with large capacity (e.g. high degree polynomial), and keep the variance in check by choosing appropriate regularization strength λ .



Bias-variance tradeoff

- Bias-variance tradeoff in the case of unregularized least squares regression ($\lambda = 0$)

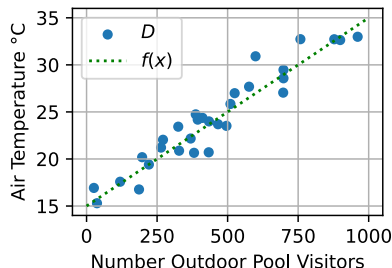


The upper-left figure from: <https://eissanematollahi.com/wp-content/uploads/2018/09/Machine-Learning-Basics-1.pdf>.

Correlation

Least squares fit

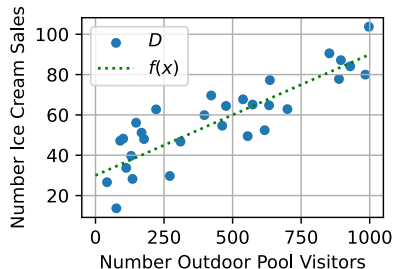
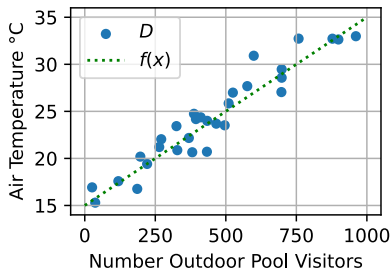
$$f(x) = 0.018x + 13.43$$



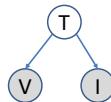
- The weights w_i can be interpreted as the strength of the (linear) relationship between feature x_i and y
- A weight of 0.018 shows a strong correlation (considering the different scales)
 - With actual data, you would normalize the data to handle the different scales of X and y and find a weight of about 1

Correlation vs. Causation

Correlation does not imply causation! Putting more people in the pool does not increase the air temperature.



Be aware of Confounding Variables!



Section 2

Probabilistic Linear Regression

In the following section, we will use probabilistic graphical models. If you do not know them yet, watch our separate Introduction to PGMs video.

Probabilistic formulation of linear regression

Remember from our problem definition at the start of the lecture,

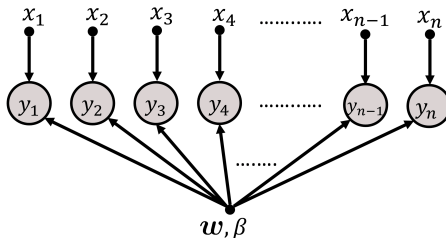
$$y_i = f_{\mathbf{w}}(\mathbf{x}_i) + \underbrace{\epsilon_i}_{\text{noise}}$$

Noise has zero-mean Gaussian distribution with a fixed precision $\beta = \frac{1}{\sigma^2}$

$$\epsilon_i \sim \mathcal{N}(0, \beta^{-1})$$

This implies that the distribution of the targets is

$$y_i \sim \mathcal{N}(f_{\mathbf{w}}(\mathbf{x}_i), \beta^{-1})$$



Remember: any function can be represented as $f_{\mathbf{w}}(\mathbf{x}_i) = \mathbf{w}^T \phi(\mathbf{x}_i)$

Maximum likelihood

Likelihood of a single sample

$$p(y_i | f_{\mathbf{w}}(\mathbf{x}_i), \beta) = \mathcal{N}(y_i | f_{\mathbf{w}}(\mathbf{x}_i), \beta^{-1}) \quad (25)$$

Assume that the samples are drawn independently

$\implies$ likelihood of the entire dataset is

$$p(\mathbf{y} | \mathbf{X}, \mathbf{w}, \beta) = \prod_{i=1}^N p(y_i | f_{\mathbf{w}}(\mathbf{x}_i), \beta) \quad (26)$$

We can now use the same approach we used in previous lecture - maximize the likelihood w.r.t. $\mathbf{w}$ and β

$$\mathbf{w}_{\text{ML}}, \beta_{\text{ML}} = \arg \max_{\mathbf{w}, \beta} p(\mathbf{y} | \mathbf{X}, \mathbf{w}, \beta) \quad (27)$$

Maximum likelihood

Like in the coin flip example, we can make a few simplifications

$$\mathbf{w}_{\text{ML}}, \beta_{\text{ML}} = \arg \max_{\mathbf{w}, \beta} p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) \quad (28)$$

$$= \arg \max_{\mathbf{w}, \beta} \ln p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) \quad (29)$$

$$= \arg \min_{\mathbf{w}, \beta} -\ln p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) \quad (30)$$

Let's denote this quantity as **maximum likelihood error function** that we need to minimize

$$E_{\text{ML}}(\mathbf{w}, \beta) = -\ln p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) \quad (31)$$

Maximum likelihood

Simplify the error function

$$E_{\text{ML}}(\mathbf{w}, \beta) = -\ln \left[\prod_{i=1}^N \mathcal{N}(y_i \mid f_{\mathbf{w}}(\mathbf{x}_i), \beta^{-1}) \right] \quad (32)$$

$$= -\ln \left[\prod_{i=1}^N \sqrt{\frac{\beta}{2\pi}} \exp \left(-\frac{\beta}{2} (\mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i) - y_i)^2 \right) \right] \quad (33)$$

$$= -\sum_{i=1}^N \ln \left[\sqrt{\frac{\beta}{2\pi}} \exp \left(-\frac{\beta}{2} (\mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i) - y_i)^2 \right) \right] \quad (34)$$

$$= \frac{\beta}{2} \sum_{i=1}^N (\mathbf{w}^T \boldsymbol{\phi}(\mathbf{x}_i) - y_i)^2 - \frac{N}{2} \ln \beta + \frac{N}{2} \ln 2\pi \quad (35)$$

Optimizing log-likelihood w.r.t. $\mathbf{w}$

$$\mathbf{w}_{\text{ML}} = \arg \min_{\mathbf{w}} E_{\text{ML}}(\mathbf{w}, \beta) \quad (36)$$

$$= \arg \min_{\mathbf{w}} \left[\frac{\beta}{2} \sum_{i=1}^N (\mathbf{w}^T \phi(\mathbf{x}_i) - y_i)^2 - \underbrace{\frac{N}{2} \ln \beta + \frac{N}{2} \ln 2\pi}_{= \text{const}} \right] \quad (37)$$

$$= \arg \min_{\mathbf{w}} \underbrace{\frac{1}{2} \sum_{i=1}^N (\mathbf{w}^T \phi(\mathbf{x}_i) - y_i)^2}_{\text{least squares error fn!}} \quad (38)$$

$$= \arg \min_{\mathbf{w}} E_{\text{LS}}(\mathbf{w}) \quad (39)$$

Optimizing log-likelihood w.r.t. $\mathbf{w}$

$$\mathbf{w}_{\text{ML}} = \arg \min_{\mathbf{w}} E_{\text{ML}}(\mathbf{w}, \beta) \quad (36)$$

$$= \arg \min_{\mathbf{w}} \left[\frac{\beta}{2} \sum_{i=1}^N (\mathbf{w}^T \phi(\mathbf{x}_i) - y_i)^2 - \underbrace{\frac{N}{2} \ln \beta + \frac{N}{2} \ln 2\pi}_{= \text{const}} \right] \quad (37)$$

$$= \arg \min_{\mathbf{w}} \underbrace{\frac{1}{2} \sum_{i=1}^N (\mathbf{w}^T \phi(\mathbf{x}_i) - y_i)^2}_{\text{least squares error fn!}} \quad (38)$$

$$= \arg \min_{\mathbf{w}} E_{\text{LS}}(\mathbf{w}) \quad (39)$$

Maximizing the likelihood is equivalent to minimizing the least squares error function!

$$\mathbf{w}_{\text{ML}} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y} = \Phi^\dagger \mathbf{y} \quad (40)$$

Optimizing log-likelihood w.r.t. β

Plug in the estimate for $\mathbf{w}$ and minimize w.r.t. β

$$\beta_{\text{ML}} = \arg \min_{\beta} E_{\text{ML}}(\mathbf{w}_{\text{ML}}, \beta) \quad (41)$$

$$= \arg \min_{\beta} \left[\frac{\beta}{2} \sum_{i=1}^N (\mathbf{w}_{\text{ML}}^T \phi(\mathbf{x}_i) - y_i)^2 - \frac{N}{2} \ln \beta + \frac{N}{2} \ln 2\pi \right] \quad (42)$$

Take derivative w.r.t. β and set it to zero

$$\frac{\partial}{\partial \beta} E_{\text{ML}}(\mathbf{w}_{\text{ML}}, \beta) = \frac{1}{2} \sum_{i=1}^N (\mathbf{w}_{\text{ML}}^T \phi(\mathbf{x}_i) - y_i)^2 - \frac{N}{2\beta} \stackrel{!}{=} 0 \quad (43)$$

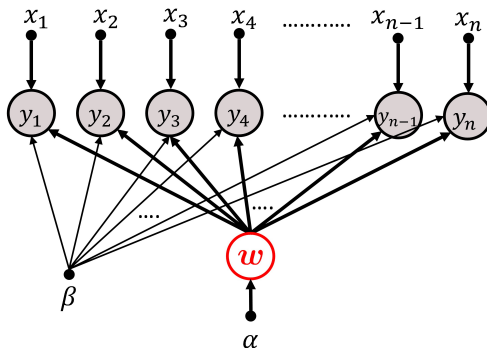
Solving for β

$$\frac{1}{\beta_{\text{ML}}} = \frac{1}{N} \sum_{i=1}^N (\mathbf{w}_{\text{ML}}^T \phi(\mathbf{x}_i) - y_i)^2 \quad (44)$$

Posterior distribution

Recall from the Lecture 3, that the MLE leads to overfitting (especially, when little training data is available).

Solution - consider the **posterior distribution** instead



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Solution - consider the **posterior distribution** instead

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}, \beta, \cdot) = \frac{\overbrace{p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta)}^{\text{likelihood}} \cdot \overbrace{p(\mathbf{w} \mid \cdot)}^{\text{prior}}}{\underbrace{p(\mathbf{y} \mid \mathbf{X}, \beta, \cdot)}_{\text{normalizing constant}}} \quad (45)$$

$$\propto p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) \cdot p(\mathbf{w} \mid \cdot) \quad (46)$$

Precision $\beta = 1/\sigma^2$ is treated as a known parameter to simplify the calculations.

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Solution - consider the **posterior distribution** instead

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$$\propto p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) \cdot p(\mathbf{w} \mid \cdot) \quad (46)$$

Connection to the coin flip example

	train data	likelihood	prior	posterior
coin:	$\mathcal{D} = \mathbf{X}$	$p(\mathbf{X} \mid \theta)$	$p(\theta \mid a, b)$	$p(\theta \mid \mathbf{X})$
regr.:	$\mathcal{D} = \{\mathbf{X}, \mathbf{y}\}$	$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta)$	$p(\mathbf{w} \mid \cdot)$	$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}, \beta, \cdot)$

How do we choose the prior $p(\mathbf{w} \mid \cdot)$?

Precision $\beta = 1/\sigma^2$ is treated as a known parameter to simplify the calculations.

Prior for $\mathbf{w}$

We set the prior over $\mathbf{w}$ to an isotropic multivariate normal distribution with zero mean

$$p(\mathbf{w} \mid \alpha) = \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \alpha^{-1} \mathbf{I}) = \left(\frac{\alpha}{2\pi} \right)^{\frac{M}{2}} \exp \left(-\frac{\alpha}{2} \mathbf{w}^T \mathbf{w} \right) \quad (47)$$

where,

α - precision of the distribution

M - number of elements in the vector $\mathbf{w}$

Motivation:

- Higher probability is assigned to small values of $\mathbf{w} \implies$ prevents overfitting (recall slide 20)
- Likelihood is also Gaussian - simplified calculations

Maximum a posteriori (MAP)

We are looking for $\mathbf{w}$ that corresponds to the mode of the posterior

$$\mathbf{w}_{\text{MAP}} = \arg \max_{\mathbf{w}} p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}, \alpha, \beta) \quad (48)$$

$$= \arg \max_{\mathbf{w}} \ln p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) + \ln p(\mathbf{w} \mid \alpha) - \underbrace{\ln p(\mathbf{y} \mid \mathbf{X}, \beta, \alpha)}_{=\text{const}} \quad (49)$$

$$= \arg \min_{\mathbf{w}} -\ln p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) - \ln p(\mathbf{w} \mid \alpha) \quad (50)$$

Similar to ML, define the MAP error function based on negative log-posterior

$$E_{\text{MAP}}(\mathbf{w}) = -\ln p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) - \ln p(\mathbf{w} \mid \alpha) \quad (51)$$

We ignore the constant terms in the error function, as they are independent of $\mathbf{w}$

MAP error function

Simplify the error function

$$\begin{aligned}E_{MAP}(\mathbf{w}) &= -\ln p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) - \ln p(\mathbf{w} \mid \alpha) \\&= \frac{\beta}{2} \sum_{i=1}^N (\mathbf{w}^T \phi(\mathbf{x}_i) - y_i)^2 - \frac{N}{2} \ln \beta + \frac{N}{2} \ln 2\pi \\&\quad - \frac{M}{2} \ln \left(\frac{\alpha}{2\pi} \right) + \frac{\alpha}{2} \mathbf{w}^T \mathbf{w} \\&= \frac{\beta}{2} \sum_{i=1}^N (\mathbf{w}^T \phi(\mathbf{x}_i) - y_i)^2 + \frac{\alpha}{2} \|\mathbf{w}\|_2^2 + \text{const} \\&\propto \underbrace{\frac{1}{2} \sum_{i=1}^N (\mathbf{w}^T \phi(\mathbf{x}_i) - y_i)^2 + \frac{\lambda}{2} \|\mathbf{w}\|_2^2}_{\text{ridge regression error fn!}} + \text{const} \quad \text{where } \lambda = \frac{\alpha}{\beta} \\&\propto E_{\text{ridge}}(\mathbf{w}) + \text{const}\end{aligned} \tag{52}$$

MAP estimation with Gaussian prior is equivalent to ridge regression!

Full Bayesian approach

Instead of representing $p(\mathbf{w} \mid \mathcal{D})$ with the point estimate $\mathbf{w}_{\text{MAP}}$, we can compute the full posterior distribution

$$p(\mathbf{w} \mid \mathcal{D}) \propto p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \beta) p(\mathbf{w} \mid \alpha). \quad (53)$$

Since both likelihood and prior are Gaussian, the posterior is as well!⁴

$$p(\mathbf{w} \mid \mathcal{D}) = \mathcal{N}(\mathbf{w} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$$

where $\boldsymbol{\mu} = \beta \boldsymbol{\Sigma} \boldsymbol{\Phi}^T \mathbf{y}$ and $\boldsymbol{\Sigma}^{-1} = \alpha \mathbf{I} + \beta \boldsymbol{\Phi}^T \boldsymbol{\Phi}$.

Observations

- The posterior is Gaussian, so its mode is the mean and $\mathbf{w}_{\text{MAP}} = \boldsymbol{\mu}$
- In the limit of an infinitely broad prior $\alpha \rightarrow 0$, $\mathbf{w}_{\text{MAP}} \rightarrow \mathbf{w}_{\text{ML}}$
- For $N = 0$, i.e. no data points, the posterior equals the prior
- Even though we assume an isotropic prior $p(\mathbf{w})$, the posterior covariance is in general not diagonal

⁴The Gaussian distribution is a *conjugate prior* of itself

Predicting for new data: MLE and MAP

After observing data $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, we can compute the MLE/MAP.

Usually, what we are actually interested in is the prediction $\hat{y}_{new}$ for a new data point $\mathbf{x}_{new}$ - the model parameters $\mathbf{w}$ are just a means to achieve this.

Recall, that we assume β to be known a priori (for simplified calculations).

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Usually, what we are actually interested in is the prediction $\hat{y}_{new}$ for a new data point $\mathbf{x}_{new}$ - the model parameters $\mathbf{w}$ are just a means to achieve this.

Recall, that $y \sim \mathcal{N}(f_{\mathbf{w}}(\mathbf{x}), \beta^{-1})$

Plugging in the estimated parameters we get a **predictive distribution** that lets us make prediction $\hat{y}_{new}$ for new data $\mathbf{x}_{new}$.

- Maximum likelihood: $\mathbf{w}_{ML}$ and β_{ML}

$$p(\hat{y}_{new} \mid \mathbf{x}_{new}, \mathbf{w}_{ML}, \beta_{ML}) = \mathcal{N}(\hat{y}_{new} \mid \mathbf{w}_{ML}^T \phi(\mathbf{x}_{new}), \beta_{ML}^{-1}) \quad (54)$$

- Maximum a posteriori: $\mathbf{w}_{MAP}$

$$p(\hat{y}_{new} \mid \mathbf{x}_{new}, \mathbf{w}_{MAP}, \beta) = \mathcal{N}(\hat{y}_{new} \mid \mathbf{w}_{MAP}^T \phi(\mathbf{x}_{new}), \beta^{-1}) \quad (55)$$

Recall, that we assume β to be known a priori (for simplified calculations).

Posterior predictive distribution

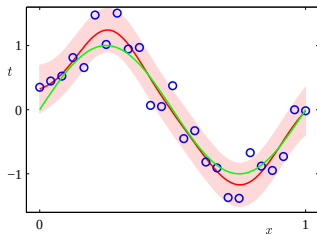
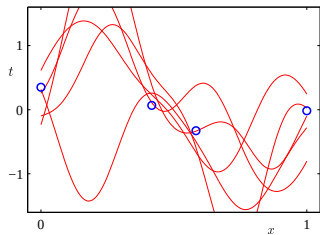
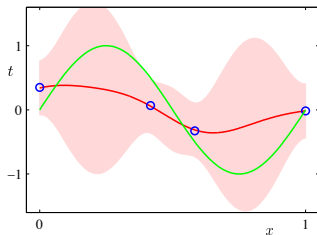
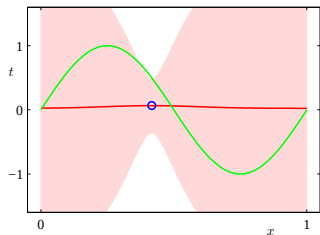
Alternatively, we can use the full posterior distribution $p(\mathbf{w} \mid \mathcal{D})$.

This allows us to compute the **posterior predictive distribution**

$$\begin{aligned} p(\hat{y}_{new} \mid \mathbf{x}_{new}, \mathcal{D}) &= \int p(\hat{y}_{new}, \mathbf{w} \mid \mathbf{x}_{new}, \mathcal{D}) d\mathbf{w} \\ &= \int p(\hat{y}_{new} \mid \mathbf{x}_{new}, \mathbf{w}) p(\mathbf{w} \mid \mathcal{D}) d\mathbf{w} \\ &= \mathcal{N}(\hat{y}_{new} \mid \boldsymbol{\mu}^T \phi(\mathbf{x}_{new}), \beta^{-1} + \phi(\mathbf{x}_{new})^T \boldsymbol{\Sigma} \phi(\mathbf{x}_{new})) \end{aligned}$$

Advantage: We get a more accurate estimate about the uncertainty in the prediction (i.e. the variance of the Gaussian, which now also depends on the input $\mathbf{x}_{new}$)

Example of posterior predictive distribution



Green: Underlying function, Blue: Observations, Dark-Red: Mode, Light-Red: Variance

Summary

- Optimization-based approaches to regression have probabilistic interpretations
 - Least squares regression $\iff$ Maximum likelihood (Slide 33)
 - Ridge regression $\iff$ Maximum a posteriori (Slide 39)
- Even nonlinear dependencies in the data can be captured by a model linear w.r.t. weights w (Slide 13)
- Penalizing large weights helps to reduce overfitting (Slide 20)
- Full Bayesian gives us data-dependent uncertainty estimates (Slide 42)

Reading material

Main reading

- “Pattern Recognition and Machine Learning” by Bishop
[ch. 1.1, 3.1, 3.2, 3.3.1, 3.3.2, 3.6]

Extra reading

- “Machine Learning: A Probabilistic Perspective” by Murphy
[ch. 7.2–7.3, 7.5.1, 7.6.1, 7.6.2]

Slides are based on an older version by G. Jensen and C. Osendorfer. Some figures are from Bishop’s “Pattern Recognition and Machine Learning”.